Machine Learning vs. Deep Learning

Customer Churn Prediction at SyriaTel and Neural Machine Translation at Google

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1. Introduction

This report compares machine learning (ML) and deep learning (DL) through two examples from the lesson. For simple machine learning, I chose customer churn prediction, which the lesson pairs with a support vector machine. For deep learning, I chose natural language processing (NLP), specifically machine translation, which the lesson pairs with recurrent neural networks (RNNs) and Transformer models. For each example I describe a documented real-world application, explain why its approach fits the problem, and explain why the other approach would be a poorer fit.

I chose these two because they sit at opposite ends of the lesson's key differences. Churn data comes from business records such as contracts, charges, and usage counts, which already fit into database tables, the kind of data I handle as a backend engineer. Translation works on raw text, where no one can list in advance the features that make a translation correct. Both real-world cases are documented by the teams that built them, so each figure in this report can be checked against a primary source.

2. Key Differences in the Lesson

The lesson defines machine learning as "a broad field in which algorithms learn from data to make predictions or decisions without being explicitly programmed" and deep learning as "a subset of machine learning that uses neural networks with many layers" (Indiana Wesleyan University, 2026). In traditional ML, people choose the features the model sees; in DL, the layers "progressively extract more abstract features from the raw input." Table 1 lists the three differences the lesson names, with the question each one raises when choosing an approach for a real problem. Sections 3 and 4 answer those questions for each example.

Table 1. *The Lesson's Key Differences and the Question Each Raises*

Difference	Machine learning	Deep learning	Question for a real project
Feature extraction	Humans often define the features manually	The algorithm learns the features from raw data	Do we know which inputs matter, and can we compute them?
Data requirements	Can work with smaller datasets and less processing power	Generally, needs large amounts of data and computing power	Do we have enough labeled examples and the hardware to train?
Complexity	Often simpler and faster to train	More layers and parameters	Do the people using the result need to see why the model decided?

Note. The first three columns paraphrase the lesson's Key Differences tab (Indiana Wesleyan University, 2026). The last column is my own framing.

The lesson's summary puts the difference as an analogy: machine learning "is like giving a student a checklist of things to look for in an email to decide if it's spam," while deep learning "is like teaching the student how to read and letting them figure out what makes an email spam based on experience" (Indiana Wesleyan University, 2026). The two examples below test that analogy. In churn prediction, the checklist can be written, and the business wants to read it. In translation, no workable checklist exists.

3. Example 1: Customer Churn Prediction (Machine Learning)

3.1 The Lesson's Example

The lesson describes a telecom company that uses "features like customer age, contract length, and monthly charges," with a support vector machine that "learns from historical data to predict whether a customer is likely to leave the service" (Indiana Wesleyan University, 2026). Each feature is chosen by a person, and each customer becomes one row of values with a yes-or-no label. That is the traditional ML pattern from the lesson's first tab: people select the features, then an algorithm learns from them (Figure 1).

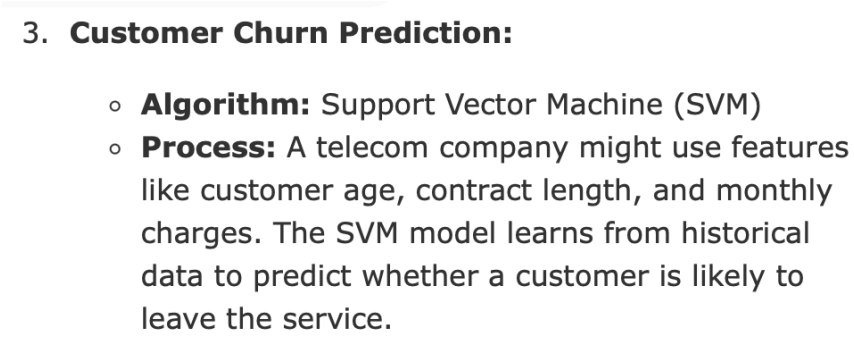


Figure 1. *The Churn Example on the Lesson's Machine Learning Examples Tab*

Note. Screenshot of the lesson cropped (Indiana Wesleyan University, 2026).

3.2 Real-World Application: SyriaTel

Ahmad et al. (2019) built a churn prediction system for SyriaTel, one of the two telecom companies in Syria. The data covered all customers over nine months: about 70 terabytes of call, SMS, internet, complaint, network, and billing records, processed on a 12-node Hadoop and Spark cluster (pp. 2, 5, 7). The team turned these records into hundreds of features per customer, including social network features such as the percentage of a customer's contacts who use the competing network (p. 10). They tested four tree-based ML algorithms. XGBoost performed best, with an area under the ROC curve (AUC) of 93.3%, followed by gradient boosting at 90.89%, random forest at 87.76%, and a single decision tree at 83% (p. 18).

The model was then tested on all 7.5 million prepaid SyriaTel customers. Marketing staff made retention offers to part of the group the model predicted would leave. About 47% of the customers who received an offer stayed, and the company's churn rate fell by about 1.5% (Ahmad et al., 2019, p. 22). SyriaTel used XGBoost rather than the support vector machine named in the lesson, but both belong to the same family: the lesson lists decision trees and support vector machines together as algorithms that learn from features people select.

3.3 Why Machine Learning Suits This Problem

The input is structured. Although SyriaTel's raw data was huge and partly unstructured, the prediction used one row of engineered features per customer, and the people who know the business can say which signals come before a customer leaves. The study confirms some of those signals: customers with many signal errors and dropped calls were among the most likely to churn (Ahmad et al., 2019, p. 21).

When the useful inputs can be named and computed, the lesson's traditional ML approach applies directly.

Human feature design improved the model. Adding the social network features the team designed raised the model's AUC from 84% to 93.3% (Ahmad et al., 2019, p. 1), a gain of 9.3 points. Feature engineering is the human step the lesson assigns to ML, and in this case it made a large, measurable difference.

The business needs the reasons. The team ranked the features by how much each one contributed to XGBoost's predictions (Ahmad et al., 2019, p. 18). A list of customers likely to leave tells a retention team whom to contact; a ranking of the reasons lets the team check the model against what it already knows about its customers. The lesson notes that traditional models are simpler, and here that simplicity is what makes the output usable.

The data is imbalanced, and boosted trees handled it. Only about 5% of SyriaTel's customers were churners (Ahmad et al., 2019, p. 6). XGBoost and gradient boosting gave their best results without any rebalancing of the data (p. 18), so the team did not need extra steps to correct for the rare class.

3.4 Why Deep Learning Is Not Suitable

Deep learning's main advantage is learning features from raw input, and that advantage shrinks when the input is already a table of meaningful columns. Grinsztajn et al. (2022) benchmarked tree-based models against deep learning on tabular data and found that "tree-based models remain state-of-the-art on medium-sized data (~10K samples) even without accounting for their superior speed" (p. 507). Among the challenges they found for neural networks was the need to be robust to uninformative features. SyriaTel's data had many of them: 77% of the numeric variables had more than 97% of their values as zero or null (Ahmad et al., 2019, p. 5). SyriaTel's 7.5 million customers are far more than 10,000 samples, so the benchmark does not settle the question by data size, but the data had the traits the benchmark found hardest for neural networks.

A deep network would also be harder to explain. The lesson notes that DL models have more layers and parameters. With many layers between the inputs and the output, the simple feature ranking SyriaTel's team reported would no longer come directly from the model, and a retention team would lose an easy way to see why a customer was flagged.

Deep learning is not impossible for churn. Ahmad et al. (2019) cite a neural network model built for about 5.23 million customers of a large Chinese telecom company that reached 91.1% accuracy (p. 3). Accuracy is a weak measure for churn, though. On data like SyriaTel's, where about 5% of customers churn, a model that predicts "stays" for every customer is about 95% accurate (100% − 5%) while finding no churners at all. That result does not show deep learning doing better than the 93.3% AUC that SyriaTel reached with XGBoost, and deep learning would add computing cost and lose transparency.

The answer would change with different data. AIMultiple lists "Churn prevention (identifying potential churners from customer feedback and behavior)" among its deep learning use cases (Dilmegani, 2026). If the main signal were free-text complaints or recorded support calls, deep learning would earn a place, at least to turn that text or audio into features. In SyriaTel's case, the signal was in records that already fit into tables.

4. Example 2: Machine Translation (Deep Learning)

4.1 The Lesson's Example

Under natural language processing, the lesson's example is machine translation, where "a deep learning model like an RNN or Transformer processes the text as a sequence of words or characters, learning context and meaning to provide accurate translations" (Indiana Wesleyan University, 2026; Figure 2). No one gives this model a list of features. It reads raw text and learns what matters.

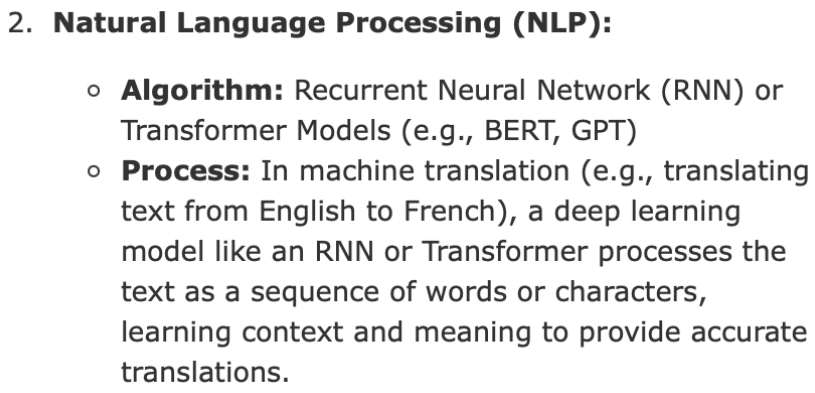


Figure 2. *The NLP Example on the Lesson's Deep Learning Examples Tab*

Note. Screenshot of the lesson, cropped (Indiana Wesleyan University, 2026).

4.2 Real-World Application: Google Neural Machine Translation

In September 2016, Google announced the Google Neural Machine Translation system (GNMT) and began using it for 100% of Chinese-to-English translations in the Google Translate mobile and web apps, "about 18 million translations per day" (Le & Schuster, 2016). GNMT is a deep long short-term memory (LSTM) network with eight encoder layers and eight decoder layers, linked by an attention mechanism (Wu et al., 2016, p. 1). An LSTM is a type of RNN, so GNMT matches the RNN option the lesson names.

In human side-by-side evaluations, GNMT "reduces translation errors by an average of 60% compared to Google's phrase-based production system" (Wu et al., 2016, p. 1). Google's announcement reports a reduction "by more than 55%-85% on several major language pairs" in sentences from Wikipedia and news sites rated by bilingual people (Le & Schuster, 2016). The two figures come from different evaluations, so they should not be combined.

4.3 Why Deep Learning Suits This Problem

Language is unstructured, and meaning depends on context. A correct translation depends on word order, idioms, agreement between words, and words whose meaning changes with the rest of the sentence. Google's earlier phrase-based system broke a sentence "into words and phrases to be translated largely independently," while the neural system "considers the entire input sentence as a unit for translation" (Le & Schuster, 2016). That is the lesson's "learning context and meaning" in practice, and it is the kind of feature discovery the lesson says deep networks do without human intervention.

Google could meet deep learning's data and computing requirements. The lesson says DL "generally requires large amounts of data and computational power." For English to French alone, the public

training set GNMT used contains 36 million sentence pairs, and training a basic model took about six days on 96 NVIDIA K80 GPUs (Wu et al., 2016, Sections 8.1 and 8.3). Serving the model at the speed Google Translate needs required Google's Tensor Processing Units (Le & Schuster, 2016). Few organizations have these resources. Google did, which is part of why deep learning was the right choice for this problem.

The neural approach removed hand-designed parts. Wu et al. (2016) write that neural machine translation "sidesteps many brittle design choices in traditional phrase-based machine translation" (p. 1), and Google's announcement says it "requires fewer engineering design choices" (Le & Schuster, 2016). This is the lesson's feature extraction difference applied to translation.

4.4 Why Traditional Machine Learning Is Not Suitable

Traditional ML needs people to define the features. In the lesson's spam example, a checklist of words and sender details works because the output is a single label. A translation is a whole new sentence in another language, and no fixed list of hand-picked features can describe how every word relates to every other word in context. Statistical, phrase-based translation "has been the dominant translation paradigm for decades" (Wu et al., 2016, p. 2), and it was the system GNMT outperformed by an average of 60% in errors.

The limits were in the hand-built pipeline. Before fully neural translation, researchers added neural networks as components inside phrase-based systems. Wu et al. (2016) note that this approach "still makes use of phrase-based translation systems at its core, and therefore inherits their shortcomings" (p. 2). Improving one component did not remove the limits of translating pieces of a sentence separately.

Deep learning here is the better choice, but it does not solve translation. Wu et al. (2016) list three weaknesses of neural translation: slower training and inference, trouble with rare words, and sometimes failing to translate every word of the source sentence (pp. 1–2). Google's announcement adds that GNMT "can still make significant errors that a human translator would never make, like dropping words and mistranslating proper names or rare terms" (Le & Schuster, 2016). The cost is the lesson's data requirement: an organization without tens of millions of example sentence pairs and large amounts of computing power would struggle to build a system like GNMT from scratch.

5. Comparing the Two Cases

Table 2 places the two cases side by side using the lesson's key differences. Data size alone did not decide either case. SyriaTel processed about 70 terabytes of raw data, yet the useful signal was in features that analysts could name, compute, and explain. Google's case turned on the form of the data: the meaning of a sentence depends on context that no feature list captures, and Google had the data and hardware to let a network learn it.

Table 2. *The Two Cases Compared*

Factor	Churn prediction (SyriaTel)	Machine translation (Google)
Lesson example	Customer churn: support vector machine on customer features	NLP: RNN or Transformer on text
Approach used	XGBoost, a tree-based ML algorithm	GNMT: 8 encoder and 8 decoder LSTM layers with attention

Factor	Churn prediction (SyriaTel)	Machine translation (Google)
Input	One row of engineered features per customer	Raw sentences
Who defines the features	The team (hundreds, including social network features)	Learned by the network
Data	Nine months of records, about 70 TB raw; 7.5 million prepaid customers tested	36 million sentence pairs for English to French alone
Computing	12-node Hadoop and Spark cluster for all processing	About 6 days on 96 NVIDIA K80 GPUs per basic model; TPUs for serving
Need to explain each result	High: a retention team needs reasons it can check	Lower: users judge the translation itself
Reported result	AUC 93.3%; about 47% of customers who got an offer stayed	Translation errors reduced by an average of 60% versus phrase-based

Note. Sources: Ahmad et al. (2019); Le and Schuster (2016); Wu et al. (2016). The "need to explain" row is my own assessment.

Figure 3 turns the comparison into the questions I would ask before choosing an approach. It follows the lesson's key differences and shows where each case lands.

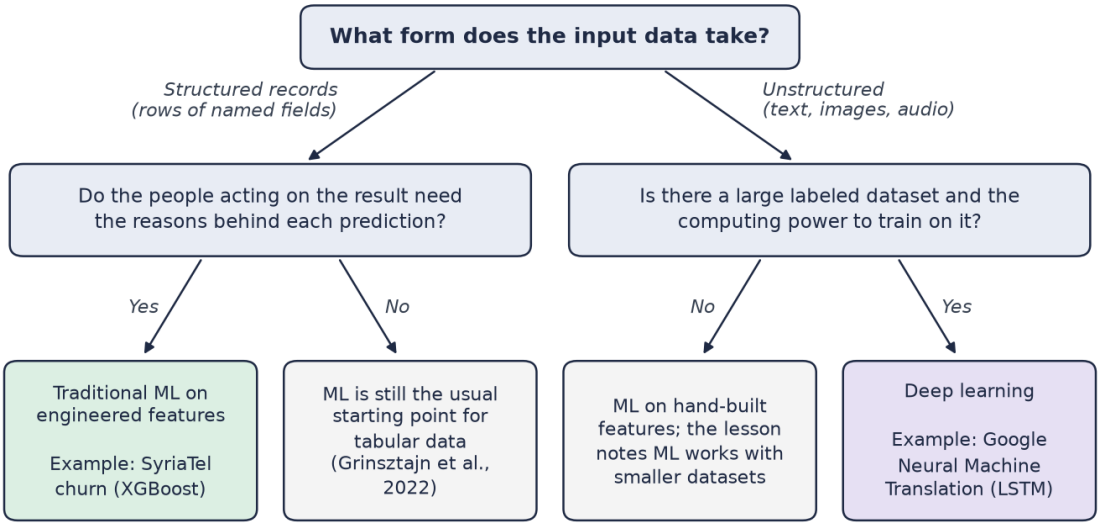


Figure 3. *Questions for Choosing Between Machine Learning and Deep Learning*

Note. Based on the lesson's key differences and the two cases in this report.

6. Conclusion

The lesson compares machine learning to giving a student a checklist and deep learning to teaching the student to read. Both cases support the comparison. SyriaTel's churn model worked because people could write the checklist, and the checklist was worth keeping: the features the team designed raised the model's AUC by 9.3 points, and the ranking of those features showed which signals predicted churn. Google's translation system worked because no checklist could capture a language, and Google had the tens of millions of examples and the computing power that deep learning needs. Deep learning is the

stronger tool only when the data and the problem call for it. When the data already sits in well-understood tables and people need to see why the model decided, traditional machine learning remains the better first choice.

Generative AI Statement

I used Claude (Anthropic) to check each figure and quotation against the original sources, to run checks such as the spell check, and to help polish and edit the wording.

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